COHEN-MACAULAY WEIGHTED CHORDAL GRAPHS

SHUAI WEI

Abstract. In this paper I give a combinatorial characterization of all the Cohen-Macaulay weighted chordal graphs. In particular, it is shown that a weighted chordal graph is Cohen-Macaulay if and only if it is unmixed.

1. Introduction

Convention. Throughout this paper, let $\mathbb{N} = \{1, 2, \ldots, \}$, $n \in \mathbb{N}$, $\mathbb{K}$ a field and $G = (V, E)$ a (finite simple) graph with vertex set $V = V(G) = [n] = \{1, \ldots, n\}$ and edge set $E = E(G)$. An edge between vertices i and j is denoted ij.

Combinatorial commutative algebra is a branch of mathematics that uses combinatorics and graph theory to understand certain algebraic constructions; it also uses algebra to understand certain objects in combinatorics and graph theory.

To the graph G one associates the positive integer-valued function $\lambda : E \to \mathbb{N}$, producing a weighted graph G_λ. For a weighted graph G_λ Paulsen and Sather-Wagstaff [3] introduce the weighted edge ideal $I(G_\lambda) \subseteq \mathbb{K}[X_1, \ldots, X_n]$ which is generated by all monomials $X_i^{\lambda(ij)} X_j^{\lambda(ij)}$ such that $ij \in E$. In particular, if λ is the constant function defined by $\lambda(ij) = 1$ for $ij \in E$, then $I(G_\lambda) = I(G)$, where $I(G)$ is the edge ideal associated to G in $\mathbb{K}[X_1, \ldots, X_n]$, given by $I(G) = (X_i X_j \mid ij \in E)$.

A weighted graph G_λ is called Cohen-Macaulay over $\mathbb{K}$ if $\mathbb{K}[X_1, \ldots, X_n]/I(G_\lambda)$ is a Cohen-Macaulay ring, and is called Cohen-Macaulay if it is Cohen-Macaulay over any field. The general problem is to classify the weighted graphs which are Cohen-Macaulay over $\mathbb{K}$. As for unweighted graphs one cannot expect a general classification theorem. Paulsen and Sather-Wagstaff [3] characterized all Cohen-Macaulay weighted K_1-corona of graphs and in particular all Cohen-Macaulay weighted trees. In this paper we classify all Cohen-Macaulay weighted chordal graphs following the classification of all Cohen-Macaulay chordal graphs by Jürgen[2]. The characterization is purely graph-theoretical, and it turns out that for weighted chordal graphs the Cohen-Macaulay property is independent of the field $\mathbb{K}$.

2. Preliminaries

In subsequent sections, let $\lambda : E \to \mathbb{N}$ be a positive integer-valued function and G_λ a weighted graph.

Definition 2.1. [1] A *path* in G is a non-empty subgraph $P = (V', E')$ of the form $V' = \{v_0, v_1, \ldots, v_r\}$ and $E' = \{v_0v_1, v_1v_2, \ldots, v_{r-1}v_r\}$, where $r \in \mathbb{N} \sqcup \{0\}$, we denote the path by $P = v_0v_1 \cdots v_r$ for simplicity and define $\mathring{P}_0 = v_1 \cdots v_r$.

Definition 2.2. [1] If $G' = (V', E')$ is a subgraph of G and $E' = \{ij \in E \mid i, j \in V'\}$, then G' is an *induced subgraph* of G; we say that V' *induces* or *spans* G' in G, and write $G' = G[V']$. A subgraph $G' = (V', E')$ of G is a *spanning* subgraph of G if V' spans all of G, i.e., if $V' = V$.

If $U \subseteq V$, we write $G - U$ for $G[V \smallsetminus U]$.

A maximal connected subgraph of G is a *component* of G. If $G_1, \ldots, G_n$ are components of G, then $G_i = G[V(G_i)]$ for $i = 1, \ldots, n$ and $V(G_1), \ldots, V(G_n)$ partition V.

Definition 2.3. A *rooted tree* T is a tree with a special vertex $r \in V(T)$ labelled as the "root" of the tree. A *rooted forest* is a graph whose components are rooted trees. A subgraph $G' = (V', E')$ of G is called a *rooted spanning forest* of G if G' is a rooted forest and V' spans G.

Definition 2.4. Let T be a rooted tree with root r. For any $v \in T \smallsetminus \{r\}$, there is a unique path from r to v, say $v_1 \ldots v_k$ with $v_1 = r$ and $v_k = v$, we say that v_i is *T-parent* of v_{i+1} and v_{i+1} is *T-child* of v_i for $i = 1, \ldots, k-1$.

If $\ell \in T$ has no T-child, we say that ℓ is a T-leaf, otherwise, it is a *T-inner vertex*. In particular, if $V(T) = \{r\}$, then r is a *T-leaf* but not a T-inner vertex.

Definition 2.5. [2] A *stable subset* or *clique* of the graph G is a subset F of $[n]$ such that $ij \in E$ for all $i, j \in F$ with $i \neq j$. We write $\Delta(G)$ for the simplicial complex on $[n]$ whose faces are the stable subsets of G. A vertex of $\Delta(G)$ is *free* if it belongs to exactly one facet of $\Delta(G)$, otherwise it is called *nonfree*.

Definition 2.6. The graph G is called *chordal* if all cycles of length > 3 has a chord. The weighted graph G_λ is called a *weighted chord graph* if the underlying graph G is chordal.

Definition 2.7. [3, Definition 1.4] A *weighted vertex cover* of G_λ is an ordered pair (V', δ') with a subset $V' \subseteq V$ and a function $\delta' : V' \to \mathbb{N}$ such that for each edge $ij \in E$, we have

(a) $i \in V'$ and $\delta'(i) \leq \lambda(ij)$, or
(b) $j \in V'$ and $\delta'(j) \leq \lambda(ij)$.

The number $\delta'(i)$ is the *weight* of v_i.

Note 2.8. If (V', δ') is a weighted vector cover of G_λ, then V' is a vector cover of G by definition.

Definition 2.9. [3, Definition 1.9] Given two weighted vertex covers (V_1', δ_1') and (V_2', δ_2') of G_λ, we write $(V_2', \delta_2') \leq (V_1', \delta_1')$ if $V_2' \subseteq V_1'$ and $\delta_2' \geq \delta_1'|_{V_2'}$. A weighted vertex cover (V', δ') is *minimal* if there does not exist another weighted vertex cover (V'', δ'') such that $(V'', \delta'') < (V', \delta')$. We define $|(V', \delta')| = |V'|$.

A weighted graph G_λ is called *unmixed* if all of the minimal weighted vertex covers of G_λ have the same cardinality, otherwise it is called *mixed*.

For the proof of our first main theorem we need the following algebraic fact:

Lemma 2.10. Let R be a Noetherian ring, $S = R[X_1, \ldots, X_n]$, $k \in \{0, \ldots, n-1\}$ and $J = (I_1X_1, \ldots, I_kX_k, \{X_i^{m_{ij}} X_j^{m_{ij}}\}_{1 \leq i < j \leq n})$ an ideal of S, where $I_1, \ldots, I_k$ are ideals of R and $m_{ij} \in \mathbb{N}$ for any $i, j \in \mathbb{N}$ such that $1 \leq i < j \leq n$. Then $X = \sum_{i=1}^n X_i$ is a non-zero divisor on S/J.

Proof. Let $A \subseteq [n]$ be nonempty. Let $(K_A)_{\lambda_A}$ be the weighted complete graph on A with the function $\lambda_A : E(K_A) \to \mathbb{N}$ given by $\lambda_A(ij) = m_{ij}$ for $i, j \in A$ such that $1 \leq i < j \leq n$. For any weighted vertex cover (V', δ') of $(K_A)_{\lambda_A}$, define an ideal $P^A(V', \delta') := (X_i^{\delta'(i)} \mid i \in V')$ in $\mathbb{K}[X_j \mid j \in A]$. Let $I((K_A)_{\lambda_A})$ be the weighted edge ideal of $(K_A)_{\lambda_A}$ in $\mathbb{K}[X_j \mid j \in A]$. Then by [3, Theorem 3.5], we have

$$I((K_A)_{\lambda_A}) = \bigcap_{\min.\ (V', \delta')} P^A(V', \delta'),$$

where the intersection is taken over all minimal weighted vertex cover of $(K_A)_{\lambda_A}$. For any $B \subseteq [n]$, let $\mathfrak{X}_B = (X_j \mid j \in B)S$. For any $T \subseteq [k]$, set $I_T = \sum_{j\in T} I_j$. Then

$$\begin{aligned} J &= (I_1X_1, \ldots, I_kX_k, I((K_{[n]})_{\lambda_{[n]}}))S \\ &= \bigcap_{T\subseteq[k]} (I_T, \mathfrak{X}_{[k]\smallsetminus T}, I((K_{[n]})_{\lambda_{[n]}}))S \\ &= \bigcap_{T\subseteq[k]} \big(I_T, \mathfrak{X}_{[k]\smallsetminus T}, I((K_{[n]\smallsetminus([k]\smallsetminus T)})_{\lambda_{[n]\smallsetminus([k]\smallsetminus T)}})\big)S \\ &= \bigcap_{\substack{T\subseteq[k] \\ \min.\ (V',\delta')}} \big(I_T, \mathfrak{X}_{[k]\smallsetminus T}, P^{[n]\smallsetminus([k]\smallsetminus T)}(V',\delta')\big)S, \end{aligned}$$

where the intersection is taken over all minimal weighted vertex cover of $(K_{[n]\smallsetminus([k]\smallsetminus T)})_{\lambda_{[n]\smallsetminus([k]\smallsetminus T)}}$. To prove that X is a non-zero divisor modulo J it suffices to show that X is a non-zero divisor modulo each of the ideals $(I_T, \mathfrak{X}_{[k]\smallsetminus T}, P^{[n]\smallsetminus([k]\smallsetminus T)}(V',\delta'))S$. It is equivalent to show that X is a non-zero divisor on

$$\frac{\overline{R}[X_1, \ldots, X_n]}{(\mathfrak{X}_{[k]\smallsetminus T}, P^{[n]\smallsetminus([k]\smallsetminus T)}(V',\delta'))},$$

where $\overline{R} = \frac{R}{I_T}$. The associated prime of the primary ideal $(\mathfrak{X}_{[k]\smallsetminus T}, P^{[n]\smallsetminus([k]\smallsetminus T)}(V',\delta'))$ generated by pure powers is $(\mathfrak{X}_{[k]\smallsetminus T}, \mathfrak{X}_{V'})$ in $\overline{R}[X_1, \ldots, X_n]$. Since (V',δ') is a minimal weighted vertex cover of the weighted complete graph $(K_{[n]\smallsetminus([k]\smallsetminus T)})_{\lambda_{[n]\smallsetminus([k]\smallsetminus T)}}$, we have by [3, Proposition 4.6] there exists a $\ell \in [n] \smallsetminus ([k] \smallsetminus T)$ but $\ell \notin V'$. Thus, X is a non-zero divisor on it. □

3. The classification

Theorem 3.1. *Let G_λ be a weighted chordal graph. Let $F_1, \ldots, F_m$ be the facets of $\Delta(G)$ which admit a free vertex satisfying that for $i = 1, \ldots, m$: there doesn't exists a rooted spanning forest $\mathfrak{F}$ of $G[F_i]$ in which each component T has a nonfree vertex ν_1^T as the root such that there is a nonfree vertex ν_0^T in $V \smallsetminus F_i$ with $\nu_0^T\nu_1^T \in E$ which satisfies that for any path $\nu_1^T v_2 \cdots v_k$ in T with v_k a T-leaf if $k \geq 2$, then $\lambda(\nu_0^T\nu_1^T) > \lambda(\nu_1^T v_2) > \lambda(v_2v_3) > \cdots > \lambda(v_{k-1}v_k)$, and which satisfies that for any component(s) S, T of $\mathfrak{F}$ if u is an S-inner vertex and v is a T-inner vertex with $u \neq v$, then*

$$\lambda(uv) > \min\{\max\{\lambda(uw) \mid u \text{ is the } S\text{-parent of } w\}, \max\{\lambda(vx) \mid v \text{ is the } T\text{-parent of } x\}\};$$

if u is an S-inner vertex such that $u\nu_0^T \in E$, then

$$\begin{aligned} \lambda(u\nu_0^T) > \min\{&\max\{\lambda(uw) \mid u \text{ is the } S\text{-parent of } w\}, \\ &\max\{\lambda(\nu_0^T\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Y\}\}; \end{aligned}$$

and if $\nu_0^S\nu_0^T \in E$, then

$$\begin{aligned} \lambda(\nu_0^S\nu_0^T) > \min\{&\max\{\lambda(\nu_0^S\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^S = \nu_0^Y\}, \\ &\max\{\lambda(\nu_0^T\nu_1^Z) \mid Z \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Z\}\}. \end{aligned}$$

Then the following conditions are equivalent.

(a) G_λ is Cohen-Macaulay;
(b) G_λ is Cohen-Macaulay over $\mathbb{K}$;
(c) G_λ is unmixed;
(d) $[n]$ is the disjoint union of $F_1, \ldots, F_m$.

Proof. (a) $\Rightarrow$ (b) is trivial.

(b) $\Rightarrow$ (c): Since $\mathbb{K}[X_1,\dots,X_n]/I(G_\lambda)$ is Cohen-Macaulay, we have $I(G_\lambda)$ is unmixed. So G_λ is unmixed by [3, Theorem 3.5 and Proposition 3.13].

(c) $\Rightarrow$ (d): Since G_λ unmixed, we have G is unmixed by [3, Proposition 1.14]. So (d) holds by [2, Theorem 2.1].

(d) $\Rightarrow$ (c): Let (V',δ') be a minimal weighted vertex cover of G_λ with $V'\subseteq[n]$ and $\delta':V'\to\mathbb{N}$. Then for $i=1,\dots,m$ we have $|V'\cap F_i|\geq|F_i|-1$ since F_i is a clique of G. Suppose for some $i\in\{1,\dots,m\}$, we have $|V'\cap F_i|=|F_i|$, i.e., $F_i\subseteq V'$. Then F_i contains a nonfree vertex by [3, Proposition 4.6]. Let $v\in F_i$. Claim. There exists a path $\ell_0\ell_1\cdots\ell_{k-1}\ell_k$ in G with $\ell_0\notin F_i$, $\ell_1,\dots,\ell_{k-1}\in F_i$ and $\ell_k=v$ satisfying that if $k\geq 2$, then $\lambda(\ell_0\ell_1)>\cdots>\lambda(\ell_{k-2}\ell_{k-1})>\lambda(\ell_{k-1}\ell_k)$.

We will use an algorithm to find such a path. Since $v\in F_i\subseteq V'$, we have there exists $w_1\in V\smallsetminus\{v\}$ such that $\delta'(v)\leq\lambda(vw_1)\underbrace{<\delta'(w_1)}_{\text{if } w_1\in V'}$. We then go through the following steps:

Step 1. Initially, let $j:=1$.

Step 2. If $w_j\notin F_i$ and $j=1$, then we have a path w_1v in G with $w_1\notin F_j$ and $v\in F_i$, so the claim is justified. If $w_j\notin F_i$ and $j\geq 2$, then by induction we have a path $w_jw_{j-1}\cdots w_1v$ in G with $w_j\notin F_i$ and $w_{j-1},\dots,w_1,v\in F_i$ such that

$$\delta'(v)\leq\lambda(vw_1)<\delta'(w_1)\leq\lambda(w_1w_2)<\delta'(w_2)\leq\cdots<\delta'(w_{j-1})\leq\lambda(w_{j-1}w_j)\underbrace{<\delta'(w_j)}_{\text{if } w_j\in V'},$$

implying that $\lambda(w_jw_{j-1})>\lambda(w_{j-1}w_{j-2})>\cdots>\lambda(w_2w_1)>\lambda(w_1v)$ and so the claim is justified. Hence in either case we jump out of the loop.

Step 3. If $w_j\in F_i$, then there exists $w_{j+1}\in V\smallsetminus\{w_j\}$ such that $\delta'(w_j)\leq\lambda(w_jw_{j+1})\underbrace{<\delta'(w_{j+1})}_{\text{if } w_{j+1}\in V'}$, so by induction there exists a path $w_{j+1}w_jw_{j-1}\dots w_1v$ with $w_j,\dots,w_1,v\in F_i$ such that

$$\delta'(v)\leq\lambda(vw_1)<\delta'(w_1)\leq\lambda(w_1w_2)<\delta'(w_2)\leq\cdots<\delta'(w_j)\leq\lambda(w_jw_{j+1})\underbrace{<\delta'(w_{j+1})}_{\text{if } w_{j+1}\in V'}.$$

Step 4. Re-define $j:=j+1$. If $w_j\notin F_i$, then go back to Step 2. If $w_j\in F_i$, then go back to Step 3. Since $|F_i|$ is finite and $w_j,\dots,w_1,v\in F_i$ are distinct to each other and F_i contains a nonfree vertex, we have after some finite loops, it will enter Step 2 and the claim will be proved.

Let $v_1\in F_i$. Then by the claim there exists a path $P_1:=\ell_0\ell_1\cdots\ell_{k-1}v_1$ in G with $\ell_0\notin F_i$ and $\ell_1,\dots,\ell_{k-1},v_1\in F_i$ satisfying that if $k\geq 2$, then $\lambda(\ell_0\ell_1)>\cdots>\lambda(\ell_{k-2}\ell_{k-1})>\lambda(\ell_{k-1}v_1)$. Assume $V(\mathring{P}_1)=F_i$. Then $k\geq 2$ since F_i contains a free vertex. So there exists a rooted spanning forest $\mathring{P}_1$ such that the unique component $\mathring{P}_1$ has a nonfree vertex $\nu_1^{\mathring{P}_1}:=\ell_1$ as the root and there is a nonfree vertex $\nu_0^{\mathring{P}_1}:=\ell_0$ in $V\smallsetminus F_i$ with $\ell_0\ell_1\in E$ which satisfies that for the unique path $\ell_1\cdots\ell_{k-1}v_1$ in the tree $\mathring{P}_1$ with v_1 a $\mathring{P}_1$-leaf if $k\geq 2$, then $\lambda(\ell_0\ell_1)>\cdots>\lambda(\ell_{k-2}\ell_{k-1})>\lambda(\ell_{k-1}v_1)$. Moreover, if ℓ_α and ℓ_β are $\mathring{P}_1$-inner vertices with $1\leq\alpha<\beta\leq k-1$, then since (V',δ') is a weighted vertex cover of G_λ and $\ell_\alpha,\ell_\beta\in V(\mathring{P}_1)=F_i\subseteq V'$, after settting $\ell_k=v_1$ we have

$$\begin{aligned}\lambda(\ell_\alpha\ell_\beta)&\geq\min\{\delta'(\ell_\alpha),\delta'(\ell_\beta)\}\\&>\min\{\lambda(\ell_\alpha\ell_{\alpha+1}),\lambda(\ell_\beta\ell_{\beta+1})\}\\&=\min\{\max\{\lambda(\ell_\alpha w)\mid\ell_\alpha\text{ is the }\mathring{P}_1\text{-parent of }w\},\max\{\lambda(\ell_\beta x)\mid\ell_\beta\text{ is the }\mathring{P}_1\text{-parent of }x\}\};\end{aligned}$$

and if ℓ_α is an $\mathring{P}_1$-inner vertex such that $u\nu_0^{\mathring{P}_1} \in E$, then $\ell_\alpha = \ell_1$, and so

$$\begin{aligned}\lambda(\ell_\alpha \nu_0^{\mathring{P}_1}) &= \lambda(\ell_0\ell_1)\\ &> \max\{\lambda(\ell_1 w) \mid \ell_1 \text{ is the } \mathring{P}_1\text{-parent of } w\}\\ &\geq \min\{\max\{\lambda(\ell_\alpha w) \mid \ell_\alpha \text{ is the } \mathring{P}_1\text{-parent of } w\}, \lambda(\nu_0^{\mathring{P}_1}\nu_1^{\mathring{P}_1})\},\end{aligned}$$

a contradiction.

On the other hand, we assume $V(\mathring{P}_1) \subsetneq F_i$. We then go through the following steps:

Step 1. Initially, let $b := 1$.

Step 2. If there exists a vertex $v \in F_i \smallsetminus (V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_b))$, then there exists a path $P_{b+1} := h_0h_1\cdots h_{k'-1}v$ in G with $h_0 \notin F_i$ and $h_1, \ldots, h_{k'-1}, v \in F_i$ satisfying that if $k' \geq 2$, then $\lambda(h_0h_1) > \cdots > \lambda(h_{k'-2}\lambda_{k'-1}) > \lambda(h_{k'-1}v)$. If $V(\mathring{P}_{b+1}) \cap (V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_b)) = \emptyset$, then we put $\mathring{P}_{b+1}$ into the rooted forest formed by $\mathring{P}_1, \ldots, \mathring{P}_b$ while making $\mathring{P}_{b+1}$ a rooted tree with root h_1 and making $\mathring{P}_{b+1}$ a component of the rooted forest. Assume $V(\mathring{P}_{b+1}) \cap (V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_b)) \neq \emptyset$. Let $t = \max\{c \geq 1 \mid h_c \in V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_b)\}$. Assume $h_t = u_s \in V(\mathring{P}_d)$ with $P_d := u_0u_1\cdots u_j$ for some $d \in \{1, \ldots, b\}$ and $s \in \{1, \ldots, j\}$. Set $P'_{b+1} := u_0u_1\cdots u_{s-1}h_th_{t+1}\cdots h_{k'-1}v$. Then P'_{b+1} is a path in G with $u_0 \notin F_i$ and $u_1, \ldots, u_{s-1}, h_t, h_{t+1}, \ldots, h_{k'-1}, v \in F_i$. Since $\lambda(u_{s-1}h_t) = \lambda(u_{s-1}u_s) \geq \delta'(u_s) = \delta'(h_t) > \lambda(h_th_{t+1})$, we have $\lambda(u_0u_1) > \cdots > \lambda(u_{s-1}h_t) > \lambda(h_th_{t+1}) > \cdots > \lambda(h_{k'-1}v)$. Re-define $P_{b+1} := P'_{b+1}$, then $\mathring{P}_{b+1}$ is merged into the component whose root is u_1, in the rooted forest formed by $\mathring{P}_1, \ldots, \mathring{P}_b$.

Step 3. If $V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_{b+1}) \subsetneq F_i$, then re-define $b := b + 1$ and go back to Step 2.

Step 4. If $V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_{b+1}) = F_i$, then there exists a rooted spanning forest $\mathfrak{F}$ such that by induction each component T in $\mathfrak{F}$, which is formed by a subset of the paths $\mathring{P}_1, \ldots, \mathring{P}_{b+1}$ say $\mathring{P}_{i_1}, \ldots, \mathring{P}_{i_\iota}$, has the nonfree vertex ν_1^T as the root, which is the first vertex of any one of the paths $\mathring{P}_{i_1}, \ldots, \mathring{P}_{i_\iota}$, such that there is a nonfree vertex ν_0^T in $V \smallsetminus F_i$ with $\nu_0^T\nu_1^T \in E$, which is the first vertex of any one of the paths $P_{i_1}, \ldots, P_{i_\iota}$, which satisfies that for the each path $\nu_1^T u_2 \cdots u_{j-1}u_j$ in the tree T with u_j a T-leaf if $j \geq 2$, then $\lambda(\nu_0^T\nu_1^T) > \lambda(\nu_1^T u_2) > \cdots > \lambda(u_{j-1}u_j)$. Moreover, for any component(s) S, T of $\mathfrak{F}$, if u is an S-inner vertex and v is a T-inner vertex with $u \neq v$, then since (V', δ') is a weighted vertex cover of G_λ and $u, v \in (V(\mathring{P}_1) \cup \cdots \cup V(\mathring{P}_{b+1})) = F_i \subseteq V'$, we have

$$\begin{aligned}\lambda(uv) &\geq \min\{\delta'(u), \delta'(v)\}\\ &> \min\{\max_{1\leq c\leq b+1}\{\lambda(uw) \mid uw \in E(\mathring{P}_c)\}, \max_{1\leq d\leq b+1}\{\lambda(vx) \mid vx \in E(\mathring{P}_d)\}\}\\ &= \min\{\max\{\lambda(uw) \mid u \text{ is the } S\text{-parent of } w\}, \max\{\lambda(vx) \mid v \text{ is the } T\text{-parent of } x\}\};\end{aligned}$$

if u is an S-inner vertex such that $u\nu_0^T \in E$, then since (V', δ') is a weighted vertex cover of G_λ,

$$\begin{aligned}\lambda(u\nu_0^T) &\geq \min\{\delta'(u), \underbrace{\delta'(\nu_0^T)}_{\text{if } \nu_0^T \in V'}\}\\ &> \min\{\max_{1\leq c\leq b+1}\{\lambda(uw) \mid uw \in E(\mathring{P}_c)\},\\ &\qquad \max\{\lambda(\nu_0^T\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Y\}\}\\ &= \min\{\max\{\lambda(uw) \mid u \text{ is the } S\text{-parent of } w\},\\ &\qquad \max\{\lambda(\nu_0^T\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Y\}\};\end{aligned}$$

and if $\nu_0^S\nu_0^T \in E$, then since (V',δ') is a weighted vertex cover of G_λ, by symmetry we assume $\nu_0^S \in V'$ and $\delta'(\nu_0^S) \le \lambda(\nu_0^S\nu_0^T)$, so

$$\begin{aligned}\lambda(\nu_0^S\nu_0^T) &\ge \delta'(\nu_0^S)\\ &> \max\{\lambda(\nu_0^S\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^S = \nu_0^Y\}\\ &\ge \min\{\max\{\lambda(\nu_0^S\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^S = \nu_0^Y\},\\ &\qquad \max\{\lambda(\nu_0^T\nu_1^Z) \mid Z \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Z\}\},\end{aligned}$$

a contradiction.

Since $|F_i|$ is finite, after some finite loops it will enter Step 4 and there will be a contradiction.

Thus, $|V' \cap F_i| = |F_i| - 1$ for $i = 1,\dots,m$. Since $[n]$ is the disjoint union $[n] = \bigcup_{i=1}^m F_i$, it follows that $|(V',\delta')| = |V'| = n - m$ and G_λ is unmixed.

(c) and (d) $\Rightarrow$ (a): We have that G_λ is unmixed and each minimal weighted vertex cover of G_λ has cardinality $n-m$. Let $S = \mathbb{K}[X_1,\dots,X_n]$. Then $\dim S/I(G_\lambda) = m$. Let $Y_i = \sum_{j\in F_i} X_j$ for $i = 1,\dots,m$. We will show that $Y_1,\dots,Y_m$ is a regular sequence on $S/I(G_\lambda)$. This then yields that G_λ is Cohen-Macaulay. Let $i \in \{2,\dots,m\}$, $F_i := \{i_1,\dots,i_\ell\}$ and assume $i_1,\dots,i_k$ are the nonfree vertices of $G[F_i]$. Then $k \in \{0,\dots,n-1\}$. Let $G' = G[[n] \smallsetminus \{i_1,\dots,i_\ell\}]$ and $\lambda' := \lambda|_{E(G')}$. Then $I(G_\lambda) = (I(G'_{\lambda'}), J_1X_{i_1},\dots,J_kX_{i_k}, J)$, where $J_j = (X_{i_j}^{\lambda(i_jr)-1}X_r^{\lambda(i_jr)} \mid i_jr \in E)$ for $j = 1,\dots,k$, and $J = (X_{i_r}^{\lambda(i_ri_s)}X_{i_s}^{\lambda(i_ri_s)} \mid 1 \le r < s \le \ell)$. Since $[n]$ is the disjoint union of $F_1,\dots,F_m$ we have all generators of the ideal $(I(G'_{\lambda'}), Y_1,\dots,Y_{i-1})$ belong to $\mathbb{K}[X_j \mid j \in [n] \smallsetminus F_i]$. Thus, if we set $R = \frac{\mathbb{K}[X_j \mid j\in[n]\smallsetminus F_i]}{(I(G'_{\lambda'}),Y_1,\dots,Y_{i-1})}$, then

$$\frac{\frac{S}{I(G_\lambda)}}{(Y_1,\dots,Y_{i-1})\frac{S}{I(G_\lambda)}} \cong \frac{R[X_{i_1},\dots,X_{i_\ell}]}{(I_1X_{i_1},\dots,I_kX_{i_k}, \{X_{i_r}^{\lambda(i_ri_s)}X_{i_s}^{\lambda(i_ri_s)} \mid 1 \le r < s \le \ell\})},$$

where for each j, the ideal I_j is the image of J_j under the residue class map onto R. Thus, Lemma 2.10 implies that Y_i is regular on $(S/I(G_\lambda))/(Y_1,\dots,Y_{i-1})(S/I(G_\lambda))$. □

Theorem 3.2. *Let G_λ be a weighted chordal graph. Let $F_1,\dots,F_m$ be the facets of $\Delta(G)$ which admit a free vertex satisfying that $[n]$ is the disjoint union of $F_1,\dots,F_m$. If G_λ is Cohen-Macaulay, then there doesn't exists a rooted spanning forest $\mathfrak{F}$ of $G[F_i]$ in which each component T has a nonfree vertex ν_1^T as the root such that there is a nonfree vertex ν_0^T in $V \smallsetminus F_i$ with $\nu_0^T\nu_1^T \in E$ which satisfies that for any path $\nu_1^Tv_2\cdots v_k$ in T with v_k a T-leaf if $k \ge 2$, then $\lambda(\nu_0^T\nu_1^T) > \lambda(\nu_1^Tv_2) > \lambda(v_2v_3) > \cdots > \lambda(v_{k-1}v_k)$, and which satisfies that for any component(s) S,T of $\mathfrak{F}$ if u is an S-inner vertex and v is a T-inner vertex with $u \ne v$, then*

$$\lambda(uv) > \min\{\max\{\lambda(uw) \mid u \text{ is the } S\text{-parent of } w\}, \max\{\lambda(vx) \mid v \text{ is the } T\text{-parent of } x\}\};$$

if u is an S-inner vertex such that $u\nu_0^T \in E$, then

$$\begin{aligned}\lambda(u\nu_0^T) > \min\{&\max\{\lambda(uw) \mid u \text{ is the } S\text{-parent of } w\},\\ &\max\{\lambda(\nu_0^T\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Y\}\};\end{aligned}$$

and if $\nu_0^S\nu_0^T \in E$, then

$$\begin{aligned}\lambda(\nu_0^S\nu_0^T) > \min\{&\max\{\lambda(\nu_0^S\nu_1^Y) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^S = \nu_0^Y\},\\ &\max\{\lambda(\nu_0^T\nu_1^Z) \mid Z \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^T = \nu_0^Z\}\}.\end{aligned}$$

Proof. Proof by contrapositive. Without loss of generality, assume there exists such a rooted spanning forest $\mathfrak{F}$ of $G[F_1]$ consisting of components $T_1, \dots, T_k$ in which each rooted tree T_i has a nonfree vertex $\nu_1^{T_i}$ as a root such that there is a nonfree vertex $\nu_0^{T_i}$ in $V \smallsetminus F_1$ with $\nu_0^{T_i}\nu_1^{T_i} \in E$ satisfying the conditions as in the statement of the theorem. Let $\mathcal{S}$ be the minimal set containing $\nu_0^{T_1}, \dots, \nu_0^{T_k}$. Without loss of generality, assume $\mathcal{S} \subseteq F_2 \sqcup \cdots \sqcup F_i$ for some $i \in \mathbb{N} \smallsetminus \{1\}$ and $F_j \cap \mathcal{S} \neq \emptyset$ for $j = 2, \dots, i$.

For $j = 2, \dots, i$, assume $F_j \cap \mathcal{S} = \{\nu_0^{T_{j_1}}, \dots, \nu_0^{T_{j_\ell}}\} =: V_j$ for some subset $\{j_1, \dots, j_\ell\} \subseteq \{1, \dots, k\}$, and define $\gamma_j : V_j \to \mathbb{N}$ by

$$\begin{aligned}\gamma_j(\nu_0^{T_{j_a}}) &= 1 + \max_{j' \in \{1,\dots,k\}} \{\lambda(\nu_0^{T_{j_a}}\nu_1^{T_{j'}}) \mid \nu_0^{T_{j_a}} = \nu_0^{T_{j'}}\} \\ &= 1 + \max_{j_{a'} \in \{j_1,\dots,j_\ell\}} \{\lambda(\nu_0^{T_{j_a}}\nu_1^{T_{j_{a'}}}) \mid \nu_0^{T_{j_a}} = \nu_0^{T_{j_{a'}}}\}.\end{aligned}$$

For $j = i+1, \dots, m$, assume $v_j \in F_j$ is a free vertex. Let

$$W := F_2 \sqcup \cdots \sqcup F_i \sqcup (F_{i+1} \smallsetminus \{v_{i+1}\}) \sqcup \cdots \sqcup (F_m \smallsetminus \{v_m\}).$$

Let $\delta : F_1 \sqcup W \to \mathbb{N}$. For $j = 2, \dots, i$, set $\delta(v) = \gamma_j(v)$ for any $v \in V_j$. Set $\delta(v) = 1$ for any $v \in W \smallsetminus (V_2 \sqcup \cdots \sqcup V_i)$. If $\nu_0^{T_{j_c}} \in V_j$ and $\nu_0^{T_{j'_d}} \in V_{j'}$ such that $\nu_0^{T_{j_c}}\nu_0^{T_{j'_d}} \in E$ for some $j, j' \in \{2, \dots, i\}$, then by assumption,

$$\begin{aligned}\lambda(\nu_0^{T_{j_c}}\nu_0^{T_{j'_d}}) &> \min\{\max\{\lambda(\nu_0^{T_{j_c}}\nu_1^{Y}) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^{T_{j_c}} = \nu_0^{Y}\}, \\ &\qquad \max\{\lambda(\nu_0^{T_{j'_d}}\nu_1^{Z}) \mid Z \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^{T_{j'_d}} = \nu_0^{Z}\}\} \\ &= \min\{\max_{j_{c'} \in \{j_1,\dots,j_\ell\}} \{\lambda(\nu_0^{T_{j_c}}\nu_1^{T_{j_{c'}}}) \mid \nu_0^{T_{j_c}}\nu_1^{T_{j_{c'}}}\}, \\ &\qquad \max_{j'_{d'} \in \{j'_1,\dots,j'_\ell\}} \{\lambda(\nu_0^{T_{j'_d}}\nu_1^{T_{j'_{d'}}}) \mid \nu_0^{T_{j'_d}} = \nu_1^{T_{j'_{d'}}}\}\} \\ &= \min\{\gamma_j(\nu_0^{T_{j_c}}) - 1, \gamma_{j'}(\nu_0^{T_{j'_d}}) - 1\} \\ &= \min\{\delta(\nu_0^{T_{j_c}}) - 1, \delta(\nu_0^{T_{j'_d}}) - 1\},\end{aligned}$$

implying that $\min\{\delta(\nu_0^{T_{j_c}}), \delta(\nu_0^{T_{j'_d}})\} \leq \lambda(\nu_0^{T_{j_c}}\nu_0^{T_{j'_d}})$. So we have $(W, \delta|_W)$ is a weighted vertex cover of $(G - F_1)_{\lambda|_{E(G-F_1)}}$. For $j = 1, \dots, k$ set $\delta(v) = 1$ for any T_j-leaf v and set $\delta(v) = 1 + \max\{\lambda(vw) \mid v \text{ is the } T_j\text{-parent of } w\}$ for any T_j-inner vertex v. Since $\mathfrak{F}$ is a spanning forest of $G[F_1]$, we have defined $\delta(v)$ for all $v \in F_1$. If u is a T_c-inner vertex and v is a T_d-inner vertex for some $c, d \in \{1, \dots, k\}$ with $c \neq d$. Then by assumption,

$$\begin{aligned}\lambda(uv) &> \min\{\max\{\lambda(uw) \mid u \text{ is the } T_c\text{-parent of } w\}, \max\{\lambda(vx) \mid v \text{ is the } T_d\text{-parent of } x\}\} \\ &= \min\{\delta(u) - 1, \delta(v) - 1\} \\ &= \min\{\delta(u), \delta(v)\} - 1,\end{aligned}$$

implying that $\min\{\delta(u), \delta(v)\} \leq \lambda(uv)$. For $j = 1, \dots, k$, we have

$$\begin{aligned}\lambda(\nu_0^{T_j}\nu_1^{T_j}) &> \max\{\lambda(\nu_1^{T_j}w) \mid \nu_1^{T_j} \text{ is the } T_j\text{-parent of } w\} \\ &= \delta(\nu_1^{T_j}) - 1,\end{aligned}$$

implying that $\delta(\nu_1^{T_j}) \leq \lambda(\nu_0^{T_j}\nu_1^{T_j})$. Furthermore, if u is an T_c-inner vertex such that $u\nu_0^{T_{j_d}} \in E$ for some $c \in \{1, \ldots, k\}$ and $j \in \{2, \ldots, i\}$, then by assumption,

$$\begin{aligned}
\lambda(u\nu_0^{T_{j_d}}) &> \min\{\max\{\lambda(uw) \mid u \text{ is the } T_c\text{-parent of } w\}, \\
&\qquad \max\{\lambda(\nu_0^{T_{j_d}}\nu_1^{Y}) \mid Y \text{ is a component of } \mathfrak{F} \text{ such that } \nu_0^{T_{j_d}} = \nu_0^{Y}\}\} \\
&= \min\{\delta(u) - 1, \max_{j_{d'} \in \{j_1, \ldots, j_\ell\}} \{\lambda(\nu_0^{T_{j_d}}\nu_1^{T_{j_{d'}}}) \mid \nu_0^{T_{j_d}}\nu_1^{T_{j_{d'}}} \in E\}\} \\
&= \min\{\delta(u) - 1, \gamma_j(\nu_0^{T_{j_d}}) - 1\} \\
&= \min\{\delta(u) - 1, \delta(\nu_0^{T_{j_d}}) - 1\},
\end{aligned}$$

implying that $\min\{\delta(u), \delta(\nu_0^{T_{j_d}})\} \leq \lambda(u\nu_0^{T_{j_d}})$. Hence $(F_1 \sqcup W, \delta)$ is a weighted vertex cover of G_λ.

For $j = 1, \ldots, k$, let $v \in F_1 \smallsetminus \{\nu_1^{T_j}\}$ and p_v its T_j-parent, since $(F_1 \sqcup W, \delta)$ is a weighted vertex cover of G_λ and

$$\begin{aligned}
\delta(p_v) &= 1 + \max\{\lambda(p_v w) \mid p_v \text{ is the } T_j\text{-parent of } w\} \\
&\geq 1 + \lambda(p_v v),
\end{aligned}$$

we have $\delta(v) \leq \lambda(p_v v)$, hence $((F_1 \sqcup W) \smallsetminus \{v\}, \delta|_{(F_1 \sqcup W) \smallsetminus \{v\}})$ is no longer a weighted vertex cover of G_λ. For $j = 1, \ldots, k$, there exists $j' \in \{2, \ldots, i\}$ and $j'_d \in \{j'_1, \ldots, j'_\ell\}$ such that $j'_d = j$, since $(F_1 \sqcup W, \delta)$ is a weighted vertex cover of G_λ and

$$\begin{aligned}
\delta(\nu_0^{T_j}) &= \delta(\nu_0^{T_{j'_d}}) \\
&= \gamma_{j'}(\nu_0^{T_{j'_d}}) \\
&= 1 + \max_{j'_{d'} \in \{j'_1, \cdots, j'_\ell\}} \{\lambda(\nu_0^{T_{j'_d}}\nu_1^{T_{j'_{d'}}}) \mid \nu_0^{T_{j'_d}}\nu_1^{T_{j'_{d'}}} \in E\} \\
&\geq 1 + \lambda(\nu_0^{T_{j'_d}}\nu_1^{T_{j'_d}}),
\end{aligned}$$

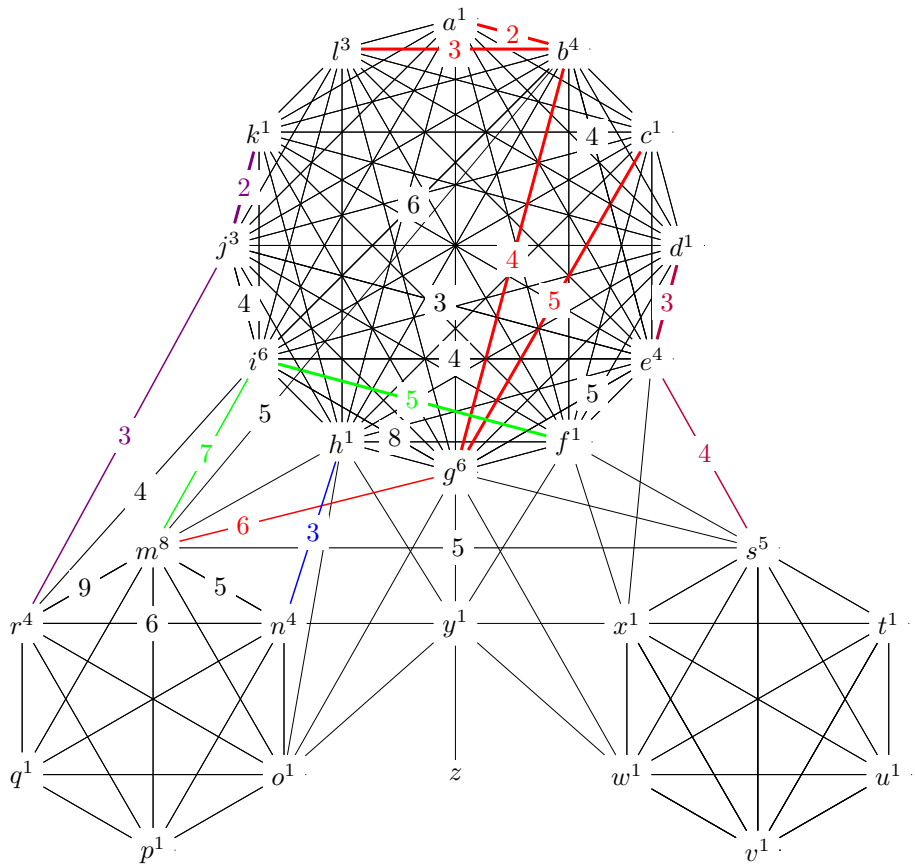
we have $\delta(\nu_1^{T_j}) = \delta(\nu_1^{T_{j'_d}}) \leq \lambda(\nu_0^{T_{j'_d}}\nu_1^{T_{j'_d}})$, hence $((F_1 \sqcup W) \smallsetminus \{\nu_1^{T_j}\}, \delta|_{(F_1 \sqcup W) \smallsetminus \{\nu_1^{T_j}\}})$ is not a weighted vertex cover of G_λ. So $((F_1 \sqcup W) \smallsetminus \{v\}, \delta|_{(F_1 \sqcup W) \smallsetminus \{v\}})$ is not a weighted vertex cover of G_λ for any $v \in F_1$. Thus, there exists a minimal weighted vertex $(F_1 \sqcup W', \delta')$ of G_λ such that $W' \subseteq W$ and $\delta' \geq \delta$ by [3, Proposition 1.12]. Since $F_2, \ldots, F_m$ are cliques of G, we have $|W'| \geq |W| - (i-1)$. So

$$|(F_1 \sqcup W', \delta')| \geq |F_1| + |W| - (i-1) = n - (m-i) - (i-1) = n - m + 1.$$

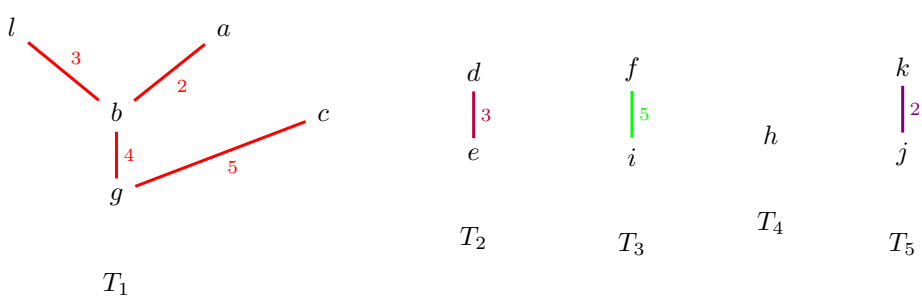
On the other hand, assume $v_j \in F_j)$ is free for $j = 1, \ldots, m$. Let $U = V \smallsetminus \{v_1, \ldots, v_m\}$ and $\delta' : U \to \mathbb{N}$ defined by $\delta'(v) = 1$. Then (U, δ') is a weighted vertex cover of G_λ. For any $v \in U$, we have $v \in F_j$ for some $j \in \{1, \ldots, m\}$ and so $vv_j \in E$, hence $(U \smallsetminus \{v\}, \delta'|_{U \smallsetminus \{v\}})$ is not a weighted vertex cover of G_λ. So there is a minimal weighted vertex cover (U, δ'') of G_λ such that $\delta'' \geq \delta'$ by [3, Proposition 1.12], but $|(U, \delta'')| = |U| = n - m$. Therefore, G_λ is mixed and so G_λ is not Cohen-Macaulay by Theorem 3.1. □

Example 3.3. The following weighted chordal graph G_λ where we only give part definition of λ is not Cohen-Macaulay. In the drawing, let ν^τ denote an element of (V', δ') with $\nu \in V'$ and

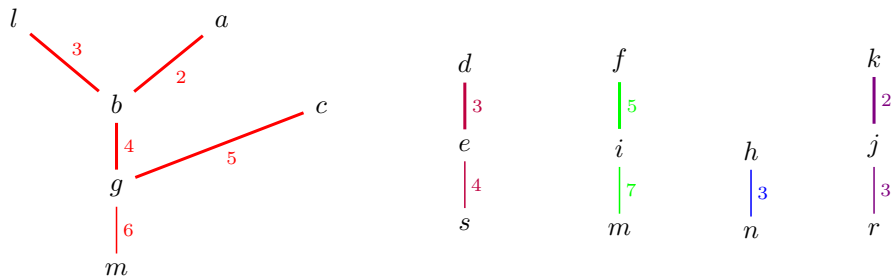
$\delta'(\nu) = \tau$.



Let $F_1 = \{a, \ldots, l\}$, $F_2 = \{m, \ldots, r\}$, $F_3 = \{s, \ldots, x\}$ and $F_4 = \{y, z\}$. There is a rooted spanning forest $\mathfrak{F}$ of $G[F_1]$ consisting of components $T_1, \ldots, T_5$ whose roots are $\nu_1^{T_1} := g, \nu_1^{T_2} := e, \nu_1^{T_3} := i, \nu_1^{T_4} = h, \nu_1^{T_5} = j$, respectively.



Let $\nu_0^{T_1} := m$, $\nu_0^{T_2} := s$, $\nu_0^{T_3} := m$, $\nu_0^{T_4} := n$ and $\nu_0^{T_5} := r$. For example, for the path gbl in T_1 with l a T_1-leaf, we have $\lambda(\nu_0^{T_1}\nu_1^{T_1}) > \lambda(\nu_1^{T_1}b) > \lambda(bl)$ since $6 > 4 > 3$.



Let $\mathcal{S} = \{m, n, s, r\}$, $i = 3$, $V_2 = F_2 \cap \mathcal{S} = \{\nu_0^{T_1} = m = \nu_0^{T_3}, \nu_0^{T_4} = n, \nu_0^{T_5} = r\}$, $V_3 = F_3 \cap S = \{\nu_0^{T_2} = s\}$. Let

$$W = F_2 \sqcup F_3 \sqcup (F_4 \smallsetminus \{z\}) = \{m, \ldots, y\}.$$

Let $\delta : F_1 \sqcup W \to \mathbb{N}$. The definition for δ is given in the drawing for G_λ. For example, since $m \in V_2$,

$$\begin{aligned}
\delta(m) &= \gamma_2(m) \\
&= 1 + \max_{j \in \{1,2,3,4,5\}} \{\lambda(m\nu_1^{T_j}) \mid m = \nu_0^{T_j}\} \\
&= 1 + \max_{j \in \{1,3,4,5\}} \{\lambda(m\nu_1^{T_j}) \mid m = \nu_0^{T_j}\} \\
&= 1 + \max\{\lambda(m\nu_1^{T_1}), \lambda(m\nu_1^{T_3})\} \\
&= 1 + \max\{\lambda(mg), \lambda(mi)\} \\
&= 1 + \max\{6, 7\} \\
&= 8,
\end{aligned}$$

For T_1-inner vertex b,

$$\begin{aligned}
\delta(b) &= 1 + \max\{\lambda(b\alpha) \mid b \text{ is the } T_1\text{-parent of } \alpha\} \\
&= 1 + \max\{\lambda(ba), \lambda(bl)\}. \\
&= 1 + \max\{2, 3\} \\
&= 4.
\end{aligned}$$

We have for example, for T_1-inner vertex b and T_3-inner vertex i,

$$\begin{aligned}
\lambda(bi) &= 6 \\
&> \min\{\max\{2, 3\}, 5\} \\
&= \min\{\max\{\lambda(ba), \lambda(bl)\}, \max\{\lambda(if)\}\} \\
&= \min\{\max\{\lambda(b\alpha) \mid b \text{ is the } T_1\text{-parent of } \alpha\}, \\
&\qquad \max\{\lambda(i\beta) \mid i \text{ is the } T_3\text{-parent of } \beta\}\};
\end{aligned}$$

for T_1-inner vertex b and $b\nu_0^{T_3} = bm \in E$, we have

$$\begin{aligned}\lambda(bm) &= 5\\ &> \min\{\max\{2,3\}, \max\{6,7\}\}\\ &= \min\{\max\{\lambda(ba), \lambda(bl)\}, \max\{\lambda(mg), \lambda(mi)\}\}\\ &= \min\{\max\{\lambda(ba), \lambda(bl)\}, \max\{\lambda(m\nu_1^{T_1}), \lambda(m\nu_1^{T_3})\}\}\\ &= \min\{\max\{b\alpha \mid b \text{ is the } T_1\text{-parent of } \alpha\},\\ &\qquad \max_{j\in\{1,2,3,4,5\}}\{\lambda(m\nu_1^{T_j}) \mid m = \nu_0^{T_j}\}\};\end{aligned}$$

for $ms \in E$, we have

$$\begin{aligned}\lambda(ms) &= 5\\ &> \min\{\max\{6,7\}, \max\{4\}\}\\ &= \min\{\max\{\lambda(mg), \lambda(mi)\}, \max\{\lambda(se)\}\}\\ &= \min\{\max\{\lambda(m\nu_1^{T_1}), \lambda(m\nu_1^{T_3})\}, \max\{\lambda(s\nu_1^{T_2})\}\}\\ &= \min\{\max_{j\in\{1,2,3,4,5\}}\{\lambda(m\nu_1^{T_j}) \mid m = \nu_0^{T_j}\},\\ &\qquad \max_{j\in\{1,2,3,4,5\}}\{\lambda(s\nu_1^{T_j}) \mid s = \nu_0^{T_j}\}\}.\end{aligned}$$

One can check that the given weighted vertex cover $(F_1 \sqcup W = \{a, \ldots, y\}, \delta)$ is a weighted vertex cover of G_λ but $((F_1 \sqcup W) \smallsetminus \{v\}, \delta|_{(F_1\sqcup W)\smallsetminus\{v\}})$ is not a weighted vertex cover of G_λ for any $v \in F_1$. Hence there exists a minimal weighted vertex cover of G_λ with cardinality ≥ 23.

References

[1] Reinhard Diestel. *Graph theory*, volume 173 of *Graduate Texts in Mathematics*. Springer, Berlin, fifth edition, 2018. Paperback edition of [MR3644391].

[2] Jürgen Herzog, Takayuki Hibi, and Xinxian Zheng. Cohen-Macaulay chordal graphs. *J. Combin. Theory Ser. A*, 113(5):911–916, 2006.

[3] Chelsey Paulsen and Sean Sather-Wagstaff. Edge ideals of weighted graphs. *J. Algebra Appl.*, 12(5):1250223, 24, 2013.

Email address: `wei6@clemson.edu`

School of Mathematical and Statistical Sciences, Clemson University, E-008 Martin Hall, Box 340975, Clemson, SC 29634, USA